

Dr. Doc:

👋 Welcome to Microbiology class. 🧑🏻‍🔬❤️🌟 INFECTIVE ENDOCARDITIS (IE) 🌟🦠

🌟 “Inflammation of the Endocardial Surface of the Heart due to microbial pathogen” 🌟

❤️ Definition

🦠 Infective Endocarditis (IE)

👉 is a microbial infection involving the endocardial surface of the heart, including:

✅ Heart valves

✅ Mural endocardium

✅ Low-pressure side of a ventricular septal defect(VSD).

📝 Note

📌 The mural endocardium is the inner lining of the heart's chambers (atria and ventricles) that is not part of the heart valves. While valvular endocarditis is more common, infections or damage to the mural endocardium-known as mural endocarditis -can lead to serious conditions like blood clots, strokes, and embolisms.

💡 The hallmark lesion is the VEGETATION 🌱

🌱 Vegetation Consists of:

🩸 Platelets

🧵 Fibrin

🦠 Microcolonies of microorganisms

🕒 Scant inflammatory cells

📖 I. CLASSIFICATION

🕒 A. Based on Time Course & Severity

🔥 Acute IE

- ⚡ Rapidly destructive
 - 🦠 High virulence organisms
 - 👉 Commonly: Staphylococcus aureus.
 - ❤️ Usually affects normal valves
 - 🕒 Fatal within days–6 weeks if untreated 💀
-

🐢 Subacute IE (SABE)

- 📈 Slowly progressive over weeks–months
 - 🦠 Low virulence organisms
 - 👉 Usually Viridans streptococci
 - 💧❤️ Occurs on previously damaged valves.
-

🕒 Chronic IE

- 📅 Rare, Develops over >3 months
 - 🦠 Often due to:
 - 🍄 Fungi
 - Coxiella burnetii
-

- ❤️ B. Based on Valve Type
 - ♦ Native Valve IE (NVIE)
 - ♦ Prosthetic Valve IE (PVE)
-

👥 C. Based on Patient Category

- 💉 IV Drug Users (IVDU)

Healthcare-associated (Nosocomial) IE

II. PATHOGENESIS & PATHOPHYSIOLOGY

Stepwise Development of IE

1) Predisposing Risk Factors

 Important Risk Factors:

 Rheumatic heart disease (RHD)

 YYYYY

 Mitral regurgitation

 Aortic stenosis

 Ventricular septal defect (VSD)

 Prosthetic valves

 IV catheters

 Poor dental hygiene

 Congenital heart disease

2) Endothelial Injury

 Valvular endothelial damage occurs from:

 Turbulent blood flow

 Catheters

 Leads to deposition of:

 Platelets + Fibrin

 Forms:

 NBTE = Non-Bacterial Thrombotic Endocarditis

3 Transient Bacteremia

 Microorganisms enter bloodstream via:

 Dental procedures

 Skin breaks

 IV drug use

 Bacteria adhere to NBTE

4 Colonization & Vegetation Formation

 Adherent Bacteria multiply rapidly

 Become Encased within fibrin & platelets, protecting them from hosts defenses like phagocytosis.

 Protected from:  Phagocytosis

 Host immune defenses

5 Local & Metastatic Spread

 Valve destruction → Heart failure

 Vegetation fragments break off → Septic emboli

 Embolization Sites:  Brain

 Kidneys

 Lungs

 Spleen

III. ETIOLOGICAL AGENTS

 Main Organisms

Approximately 85–90% caused by:

 Staphylococci

 Streptococci

 Enterococci

 Native Valve IE

 Staphylococcus aureus

 Most Common Overall

 Especially in Healthcare-associated IE

   Viridans streptococci Community Acquired cause

 Prosthetic Valve Endocarditis (PVE)

 Early PVE (<12 months)

 Nosocomial infection

 Common organisms:  S. epidermidis

 S. aureus

 Late PVE (>12 months)

 Typically Community- acquired

 Common organisms:  Oral streptococci

 Coagulase-negative staphylococci (CoNS)

 Why is *S. epidermidis* Important?

 Produces BIOFILM / SLIME LAYER (PIA)

 Helps adherence to prosthetic devices 

 IV Drug Users (IVDU)

 Usually affects:

 TRICUSPID VALVE (Right side)

 Dominant organism:  *S. aureus*

 Left sided infection can involve varied etiology including

 *Pseudomonas aeruginosa*

 *Candida* spp.

 Culture Negative IE (5–10%)

Caused by:  Prior antibiotic use

 Fastidious organisms

 HACEK Group (Haemophilus, Aggregatibacter, Cardiobacterium, Eikenella, Kingella)

Other Important Causes

 *Coxiella burnetii* → Q fever

 Other Important Organisms

 *S. gallolyticus*

( Associated with colorectal cancer)

 Fungi: (*Candida*, *Aspergillus*)

IV. CLINICAL MANIFESTATIONS

Cardiac Manifestations

 New/worsening regurgitant murmur

 Valve destruction

 Congestive heart failure

 Peripheral & Extracardiac Signs

 Janeway Lesions

 Small painless hemorrhagic macules

 Palms & soles

 Osler Nodes

 Painful raised nodules

 Fingertips/toes

 Roth Spots

 Retinal hemorrhages with pale centers

 Splinter Hemorrhages

 Linear hemorrhages under nails

Other Findings

 Petechiae

 Splenomegaly

 Clubbing

 Neurological deficits

 Laboratory Findings

 Anemia

 Leukocytosis

 Microscopic hematuria

 ESR

 CRP

 Rheumatoid factor

 V. MODIFIED DUKE CRITERIA

 Definite IE Diagnosis

 2 Major Criteria

OR

 1 Major + 3 Minor

OR

 5 Minor Criteria

 Major Criteria

 Positive Blood Culture

 Typical IE organisms isolated from:

 Separate blood cultures

 Persistent positive cultures

② Evidence of Endocardial Involvement

 Positive echocardiography showing:

 Vegetation

 Abscess

 Prosthetic valve dehiscence

 Minor Criteria

 Predisposing heart disease/IVDU

 Fever $\geq 38^{\circ}\text{C}$

 Vascular phenomena

 Immunologic phenomena

 Microbiologic evidence not meeting major criteria

VI. DIAGNOSTIC WORKUP

 Blood Culture

 MOST IMPORTANT TEST 

Obtain 2–3 sets before antibiotics

 Different venipuncture sites

Timing:

 Acute IE $\rightarrow$ within 1 hour

 Subacute IE $\rightarrow$ over 24–72 hrs

 Imaging

Echocardiography

 TTE (Transthoracic Echo)

 TEE (Transesophageal Echo)

Detects:  Vegetations

 Abscesses

 Valve damage

 Non-Culture Tests

 Serology:

✓ Coxiella

✓ Brucella

 PCR: ✓ Detects microbial DNA from vegetations

 VII. TREATMENT PRINCIPLES

 Core Principles

✓ Bactericidal therapy

✓ IV  administration

Maximal doses 

✓ Long duration (4–6 weeks)

 Treatment Regimens

 S. aureus

 Native Valve

 Cloxacillin/Nafcillin (MSSA)

 Vancomycin (MRSA)

 Duration: 6 weeks

 Prosthetic Valve

+ Rifampin (6 weeks)

+ Gentamicin (2 weeks)

🌿 Viridans Streptococci

💊 Penicillin or Ceftriaxone

⌚ 4 weeks (native valve)

+ Gentamicin in prosthetic valve cases

🧫 HACEK Organisms

💊 Ceftriaxone

OR

💊 Ciprofloxacin

⌚ 4 weeks (native)

⌚ 6 weeks (prosthetic)

🔵 Enterococci

💊 High-dose Penicillin/Ampicillin

+ Gentamicin

⌚ 6 weeks

🧪 Therapeutic Drug Monitoring

💊 Vancomycin

Target trough:

💊 Gentamicin

Peak should remain:

⚠️ Prevents nephrotoxicity & ototoxicity

⚠️ VIII. COMPLICATIONS

❤️ Cardiac Complications

💔 Congestive heart failure (MOST COMMON)

❤️ Myocardial abscess

👁️ Cardiac fistulas

🌐 Extracardiac Complications

🧠 Embolic stroke

🧠 Encephalopathy

🩺 Renal failure

🧬 Immune complex glomerulonephritis

🩸 Mycotic aneurysm

🧠 HIGH-YIELD PEARLS ✨

☀️ S. aureus → Most common overall cause

☀️ IVDU → Usually affects TRICUSPID valve

☀️ S. epidermidis → Prosthetic valve + biofilm formation

☀️ Blood culture before antibiotics 🧪

☀️ Vegetation = hallmark lesion 🌱

☀️ CHF = most common complication 💔

☀️ Duke Criteria = cornerstone of diagnosis 📖

🏆 QUICK MEMORY TRICK

🧠 “FROM JANE TO OSLER”

👉 Janeway = Painless

👉 Osler = Painful

✨ “Jane has NO pain, Osler is OUCH!” 😊

🦠 ENDOCARDITIS = “DAMAGED VALVE + BACTEREMIA + VEGETATION” 🦠🦠

❤️🔥 MYOCARDITIS 🔥❤️

🦠 Inflammation of the Myocardium 🦠

📖 Definition

🦠 Myocarditis is an inflammatory disease involving the MYOCARDIUM ❤️ (the thick, contractile muscular layer of the heart).

⚠️ It is one of the three major cardiovascular infections affecting the layers of the heart.

💡 Clinical 🩺 presentation ranges from: 😊 Mild/subclinical disease ➡️ to 💀 Fulminant fatal heart failure

✓ Presents with a wide range of clinical manifestations 🩺

🦠 I. ETIOLOGY

☀️ Causes of Myocarditis

📖 The causes are highly diverse, ranging from infectious organisms 🦠 to Autoimmune disorders 🧑🏻🩺, & Toxins 💊🚫.

🧑🏻🩺 Autoimmune diseases

💊🚫 Drugs & toxins

🦠 Viral infections

🦠 Bacterial infections

🦠 Parasites

🦠 Fungi

A. VIRAL CAUSES (MOST COMMON)

▲ Viruses  are the most common cause of Myocarditis.

 Coxsackie B virus is the most frequently implicated agent ★ Most Common Agent

 Coxsackie B Virus

 Other Viral Causes

 Coxsackie A

 Enteroviruses

 Echovirus

 Influenza A & B

 HIV

 Adenovirus

 Parvovirus B19

 Mumps

 Rubella

 Measles

 RSV (Respiratory Syncytial Virus)

 Varicella-Zoster Virus (VZV)

 Herpesviruses

 CMV

 EBV

 HHV-6

B. BACTERIAL CAUSES

Important Bacteria

 *Corynebacterium diphtheriae*  diphtheriae   Diphtheria toxin  can directly damage  myocardium

 *Borrelia burgdorferi* (the cause of  Lyme disease)

 Group A *Streptococcus*  Associated with Rheumatic Fever

Other Bacterial Causes

 *Staphylococcus aureus*

 *Neisseria meningitidis*

 *Clostridium perfringens*

 *Mycoplasma pneumoniae*

 *Chlamydia pneumoniae*

 *Chlamydia psittaci*

 Rickettsial Species

 *Rickettsia rickettsii*

 *Rickettsia tsutsugamushi*

C. PARASITIC CAUSES

★ HIGH-YIELD ORGANISM

 *Trypanosoma cruzi*

 Causes:   CHAGAS DISEASE (American Trypanosomiasis)

 Major cause of chronic dilated cardiomyopathy.

D. FUNGAL CAUSES

 Rare but important in:  Immunocompromised patients

Common Fungi

 Aspergillus

 Candida

 E. NON-INFECTIOUS / AUTOIMMUNE CAUSES

 Acute Rheumatic Fever (ARF)

 Causes:  PANCARDITIS (Inflammation  of ALL heart layers)

 Mechanism: Molecular Mimicry

 YYY-Antibodies against streptococcal M protein  cross-react with:

 Cardiac myosin

 Autoimmune myocardial injury.

Other Autoimmune Causes

 SLE (Systemic Lupus Erythematosus)

 Rheumatoid Arthritis

 Allergic reactions

 TOXINS & DRUGS

 Important toxic causes:

 Anthracyclines

 Cocaine

 Various cardiotoxic drugs

 II. PATHOGENESIS & SPREAD

Routes of Spread

 Microorganisms can reach the myocardium through several routes:

 Hematogenous Spread (MOST COMMON)

 Via: Viremia

Bacteremia

 Contiguous Spread

Spread from nearby infected organs:  Especially lungs

 Lymphangitic Spread

Through lymphatic channels

 Direct Inoculation

Occurs through:  Surgery

 Trauma

 Viral Mechanism (Coxsackie B)

 Stepwise Pathogenesis

① Enteroviruses like Coxsackie B initiate replication in the mucosa and lymphoid tissue (pharynx/tonsils)

Viral replication in:  Pharynx

 Tonsils

 Lymphoid tissue



② Spread to:  Peyer's patches



3] And enter the bloodstream Primary viremia 🩸



4] Reaches myocardium ❤️

💀 Cytolytic Effect

🦠 Most enteroviruses are:

💥 CYTOLYTIC

➡️ meaning they directly damage target heart cells/Directly destroy myocardial cells

🩸 Pathogenesis 💀 of T. cruzi

🩸 Parasites multiply locally as AMASTIGOTES intracellularly(in myocytes)



🧫 Formation of pseudocysts



💥 Rupture of pseudocysts



🔥 Intense inflammation



💔 Tissue destruction

🩺 III. CLINICAL PRESENTATION

⚠️ Presentation ranges from: 😊 Mild illness ➡️ 💀 Fulminant myocarditis

😊 Mild Presentation

Often resembles viral infection:

-  Fever
 -  Headache
 -  Sore throat
 -  Rash
-

-  Severe Presentation
 -  Cardiac Symptoms
 -  Chest pain
 -  Dyspnea/SOB
 -  Palpitations
 -  Arrhythmias
-

-  Newborn Myocarditis
 -  MOST LIFE-THREATENING in newborns.
 - Presentation:  Sudden unexplained heart failure
 -  High mortality rate
-

-  Myopericarditis
 -  Myocarditis often coexists with:
 -  Pericarditis
 -  Concurrent inflammation is called:
 -  MYOPERICARDITIS
-

IV. COMPLICATIONS

-  Congestive Heart Failure (CHF)
Leads to:  Edema
 Breathlessness
 Hepatic congestion
 -  Cardiogenic Shock
-

💀 Severe reduction in cardiac output ➡ Life-threatening emergency

3. ⚡ Sudden Cardiac Death

⚠️ Myocarditis 🗑️ 🔥 is Responsible for up to 20% of sudden deaths in young adults 😬

💡 Often due to: ⚡ Ventricular arrhythmias/block

4. 🗑️ Chronic Dilated Cardiomyopathy

⚠️ Can occur YEARS–DECADES later

Especially with: 🩸 Untreated Chagas disease

🔬 V. DIAGNOSTIC WORKUP

🔬 Laboratory Findings

🔥 Inflammatory Markers

⬆️ WBC count

⬆️ ESR

⬆️ CRP

❤️ Cardiac Biomarkers

Markers of Myocardial Injury

⬆️ Troponin T/I

⬆️ Creatine Kinase (CK)

⬆️ Myoglobin

🧪 Microbiological Tests

🩸 Blood Cultures

Used to: ✅ To rule out ❌ /Exclude bacterial causes

🧬 Viral Studies

🔬 Viral serology

🧬 RT-PCR

⚠️ Serology is difficult because: 🦠 Many viral serotypes exist.

ECG Findings

Common ECG Abnormalities

- ⚡ ST elevation
 - ⚡ Non-specific T-wave abnormality
-

Imaging

Chest X-Ray

May show:

- ❤️ CARDIOMEGALY  (Enlarged heart)
-

Advanced Imaging

MRI

Radiology studies

- Useful for: ☒ Detecting inflammation
- ☒ Evaluating ventricular function
-

ENDOMYOCARDIAL BIOPSY

★ DEFINITIVE DIAGNOSTIC TEST ★

-  Confirms myocarditis histologically
 - ⚠️ Reserved for severe cases because:
 - ❌ Invasive
 - ❌ Risk of complications
-

HIGH-YIELD PEARLS ✨

- ☀️ Coxsackie B = MOST COMMON cause
- ☀️ Myocarditis may mimic viral flu illness 🤒

- ☀️ Troponin elevation = myocardial injury ❤️
 - ☀️ Sudden cardiac death in young adults ⚡
 - ☀️ Chagas disease → dilated cardiomyopathy 🐍
 - ☀️ Endomyocardial biopsy = GOLD STANDARD 🧬
 - ☀️ Myocarditis + Pericarditis = Myopericarditis 🔥
-

🎯 QUICK MEMORY TRICK

🧠 “COXSACKIE CRASHES THE HEART” ☀️ ❤️

🦠 Cocksackie B

➡️ Viral replication

➡️ Cytolytic myocardial injury

➡️ Arrhythmia + Heart failure

❤️ 🔥 MYOCARDITIS = “INFLAMED HEART MUSCLE WITH ARRHYTHMIA & HEART FAILURE RISK”
🔥 ❤️

🔥 🫀 PERICARDITIS 🫀 🔥

☀️ Inflammation of the Protective Sac Around the Heart ☀️

📖 Definition

🫀 Pericarditis is an inflammatory disease of the PERICARDIUM ❤️—the fibrous protective sac surrounding the heart.

✨ Classic Triad:

☀️ Chest pain

🎧 Pericardial friction rub 🎧

📊 ECG changes

I. ETIOLOGY

Causes of Pericarditis

✓ Pericarditis is commonly:

 INFECTIOUS, But may also result from:

 Autoimmune disease

 Metabolic disorders

 Idiopathic causes

A. VIRAL CAUSES (MOST COMMON)

Major Viral Cause

 Coxsackie B Virus

Less commonly:  Coxsackie A

Other Viral Causes

 Echovirus

 Parvovirus

 EBV

 HIV

 Mumps

 Influenza A & B

 Rubella

B. BACTERIAL CAUSES

HIGH-YIELD ORGANISM

 Mycobacterium tuberculosis

 Causes:

 TUBERCULOUS PERICARDITIS

Other Important Bacteria

 Streptococcus pneumoniae

 Staphylococcus aureus

 Neisseria gonorrhoeae

 Neisseria meningitidis

 Borrelia burgdorferi (Lyme disease)

 C. FUNGAL / MYCOTIC CAUSES

 Usually seen in: Immunocompromised patients

 Post-cardiac surgery patients.

Important Fungi

 Candida

 Aspergillus

 D. AUTOIMMUNE / INFLAMMATORY CAUSES

 Acute Rheumatic Fever

 Causes:

 PANCARDITIS(Inflammation of ALL heart layers)

Other Autoimmune Disorders

 SLE

 Rheumatoid Arthritis (RA)

 Scleroderma

 Reiter Syndrome

 Dermatomyositis

 E. METABOLIC CAUSES

 Uremia (renal failure)

 Hypothyroidism

 F. IDIOPATHIC

Cause remains unknown

 II. PATHOGENESIS & SPREAD

 Pathophysiology

 Inflammation causes production of:

 FIBRINOUS EXUDATE.

Resulting Changes

 Pericardium becomes:

 Dull

 Opaque

 Sandy appearing

 Eventually causes:

 Adhesion formation

 Fibrosis

 Routes of Spread

1 Contiguous Spread

From nearby organs: ➡ Especially lungs 🫁 (common in TB)

🩸 2 Hematogenous Spread

Through bloodstream: 🦠 Viremia

🦠 Bacteremia

⚠️ Common in miliary TB

🦠 3 Lymphangitic Spread

Via lymphatic channels

🔪 4 Direct Inoculation

Occurs after: 🏥 Surgery

💥 Trauma

📖 III. CLASSIFICATION

🫀 A. TYPES OF PERICARDITIS

1 Caseous Pericarditis

💀 Specifically associated with Tuberculosis

🦠 Tuberculous pericarditis

2 Serous Pericarditis

YYYY Associated with autoimmune diseases like 🦴 RA and 🦋 SLE

3 Fibrous Pericarditis

A chronic form occurring after prolonged inflammation from 🦠 TB or 🍄 Fungal infections.

4 Rheumatic Pericarditis

🎯 Part of a pancarditis, often presenting as fibrinous pericarditis.

Occurs in: 🏴‍☠️ Acute Rheumatic Fever

💧 B. TYPES OF PERICARDIAL EFFUSION

1 Serous Effusion

💧 Transudative fluid

Common in: 💔 Heart failure

2 Suppurative Effusion

🦠 Pyogenic infection

Contains: ⚪ Leukocytes

🗑️ Cellular debris

3 Hemorrhagic Effusion

🩸 Blood-containing fluid

Associated with: 🦠 Infection

🚨 Malignancy

4 Serosanguinous Effusion

💧 🩸 Mixed serous + bloody fluid

IV. CLINICAL PRESENTATION

Symptoms may range from: 😊 Mild viral-like illness ➡ to

🔥 Severe cardiac compromise

💣 CHEST PAIN (CLASSIC)

★ Sharp & Positional ★

⚠️ Worse 🤯 when: 🛌 Lying flat

✅ Relieved by: ➡ Sitting forward

PERICARDIAL FRICTION RUB

★ MOST CHARACTERISTIC SIGN ★

✚ Extra heart sound 🗣️ consisting of:

🫀 1 systolic component

🫀 2 diastolic components

✓ described as grating, scratching, or rasping.

Sound Description

🎵 Grating

🎵 Scratching

🎵 Rasping

🌡️ Other Symptoms

🌡️ Intermittent fever

 Dyspnea

 Tachypnea

 Cough

 Dysphagia

 TB-Specific Features

 Weight loss

 Night sweats

 Hepatomegaly

 V. DIAGNOSTIC WORKUP

 ECG FINDINGS

★ Classic ECG Changes

 ST Segment Elevation

Other ECG Findings

 PR Depression

 T-wave inversion (later stage)

 LABORATORY TESTS

 Inflammatory Markers

 Leukocytosis

 ESR

 CRP

 Routine Tests

 Blood cultures

 Urea

 Creatinine

 TB TESTS

 PPD Skin Test

Usually positive in TB

 Limitation:  BCG vaccine may cause false positives

 Interferon Test

★ More specific for TB

 Viral Studies

 PCR

 Viral serology

 Serology difficult because:  Many viral serotypes exist

 IMAGING

 Chest X-Ray

May reveal:  Enlarged cardiac shadow

 May show an enlarged cardiac shadow (if effusion >250mL) or a calcified pericardium.

 Calcified Pericardium

Suggestive of:  Chronic constrictive disease

 Echocardiography (TTE)

★ BEST TEST FOR EFFUSION

Detects: 💧 Pericardial fluid

💔 Tamponade physiology

🦴 CT Scan

✓ CT Scan: Can show pericardial thickening (>5mm)

📺 MRI

Useful for: 🔥 Detecting inflammation

💓 Structural evaluation

🔬 DEFINITIVE DIAGNOSIS

★ Pericardial Biopsy & Culture ★

Especially important in: 💀 Tuberculous pericarditis

💊 VI. MANAGEMENT

☀️ Viral / Idiopathic Cases

Supportive Treatment

🛌 Bed rest

💊 NSAIDs

💊 Colchicine

🦠 Specific Therapy

Bacterial Causes

💊 Antibiotics

⚠️ Must cover: 🟡 S. aureus

Viral Causes

 Antivirals

Example:  Acyclovir for herpesvirus

Tuberculous Pericarditis

 Anti-TB therapy

 Corticosteroids

 Goal: Prevent fibrosis & scarring

 Pericardiocentesis

 EMERGENCY PROCEDURE

Performed to relieve:

 CARDIAC TAMPONADE

 Pericardiectomy

May be needed for:  Constrictive pericarditis

VII. COMPLICATIONS

1.  Pericardial Effusion

Fluid accumulation around heart

2.  Cardiac Tamponade

 LIFE-THREATENING

 Compression of heart by fluid

Results in:  Reduced ventricular filling

 Shock

3. 🪨 Constrictive Pericarditis

★ DREADED LONG-TERM COMPLICATION ★

Commonly due to: 💀 TB

Pathophysiology

Pericardium becomes: 🪨 Thick

🪨 Fibrotic

🪨 Non-yielding

➡ Restricts ventricular filling

💔 Other Complications

💔 Heart failure

🫁 Pleural effusion

🫀 Restrictive pericarditis

🧠 HIGH-YIELD PEARLS ✨

☀️ Coxsackie B = common viral cause

☀️ TB = major bacterial cause worldwide 💀

☀️ Chest pain relieved by leaning forward ↘

☀️ Pericardial friction rub = hallmark sign 🎧

☀️ ST elevation + PR depression on ECG 📊

☀️ TTE = best test for effusion 💧

☀️ Tamponade = emergency 🚨

☀️ Constrictive pericarditis = severe chronic complication 🪨

QUICK MEMORY TRICK

 “PERI-CARDITIS PAIN PREFERS FORWARD” ↘  

Pain worse lying down

 Better sitting forward

  PERICARDITIS = “INFLAMED PERICARDIAL SAC CAUSING POSITIONAL CHEST PAIN & FRICTION RUB”  

  CARDIOVASCULAR INFECTIONS & PARASITIC HEART DISEASES  

 High-Yield Master Note for CVS Infectious Diseases 

I. INFECTIONS OF THE VEINS

 SEPTIC / SUPPURATIVE THROMBOPHLEBITIS (STP)

Definition

 Septic (Suppurative) Thrombophlebitis is:

 Infection + Venous Thrombosis

 Presence of an:

 ENDOVASCULAR THROMBUS

in association with:  Bacterial or fungal infection

 Etiology & Predisposing Factors.

Common causes include:

 Intravenous catheters

 Skin breakdown

 Spread from adjacent infected structures

SPECIFIC STP SYNDROMES

1] LEMIERRE'S SYNDROME

 Affected Vein:

 Internal jugular vein

 Predisposing Event

 Exudative pharyngitis

 Causative Organism

 *Fusobacterium necrophorum*

 Treatment

 IV Penicillin G

2] PYLEPHLEBITIS

 Affected Vein:

 Portal vein

 Predisposing Conditions

 Diverticulitis

 Intra-abdominal infections

 Common Organisms

 *Bacteroides fragilis*

 Intestinal flora

 Treatment

 Broad-spectrum IV antibiotics

 Heparin

3] SEPTIC PELVIC THROMBOPHLEBITIS

 Affected Veins:

 Ovarian vein & pelvic veins

 Common Scenario

 Post-partum state

 Organisms

 Intestinal flora

 Treatment

 Broad-spectrum IV therapy

 Heparin

4] LINE-RELATED SEPTIC PHLEBITIS

 Major Cause

 Indwelling IV catheters

 Common Organisms

 Staphylococci

 Treatment

 IV anti-staphylococcal antibiotics

 Excision of purulent small veins

 GENERAL MANAGEMENT OF STP

Main Principles

 High-dose antibiotics

 Possible heparin therapy

II. TUBERCULOSIS (TB) & THE HEART

 MAIN CARDIAC MANIFESTATION

 TUBERCULOUS PERICARDITIS

 Most common cardiac involvement in active TB

 PATHOPHYSIOLOGY

Inflammation produces:

 FIBRINOUS EXUDATE

 Pericardial Changes

Pericardium becomes:

 Dull

 Opaque

 “Sandy sac”

 Leads to:  Adhesion formation

 Fibrosis

 CLASSIFICATION

1 Caseous Pericarditis

★ Classic TB-associated type

2 Fibrous Pericarditis

Occurs after prolonged inflammation

 ROUTES OF SPREAD

 Contiguous Spread

From lungs or nearby organs

 Hematogenous Spread

Especially in:

⚠ Miliary TB

 CLINICAL PRESENTATION

 Fever

 Pericardial friction rub

 Weight loss

🌙 Night sweats

 Hepatomegaly

⚠ COMPLICATIONS

 Constrictive Pericarditis

★ DREADED COMPLICATION ★

Pericardium becomes:  Thick

 Fibrotic

 Non-yielding

➡ Restricts ventricular filling 

Other Complications

 Pericardial effusion

 Cardiac tamponade

 Heart failure

 DIAGNOSTIC WORKUP

 PPD SKIN TEST

Usually positive

+ ⚠ False positives occur with:  BCG vaccination

 INTERFERON TEST

★ More specific for TB

 ECG Findings

Classic Change:

 ST Segment Elevation

Additional ECG Changes

 PR depression

 T-wave inversion

 Definitive Diagnosis

★ Pericardial Biopsy & Culture ★

More definitive than fluid smear

III. SPIROCHETES & THE HEART

A. SYPHILIS

 Organism:

 Treponema pallidum

 Timing of Cardiac Disease

Occurs during:

 TERTIARY SYPHILIS

Usually many years later

 Pathogenesis

Spirochetes invade: ❤️ Aortic root

 Cause: 🔥 Destructive granulomatous inflammation

❤️ Cardiac Effects

💓 Aortic root dilation

💓 Aortic valve insufficiency

 Important Point

Perform syphilis serology in: 😬 Older adults with acquired aortic root dilation

B. LYME DISEASE

 Organisms:

 Borrelia burgdorferi

 Borrelia afzelii

 Transmission

 Tick✓ bite

 Early Skin Manifestation

 ERYTHEMA CHRONICUM MIGRANS (ECM)

 Expanding ring-like lesion

 CARDIAC MANIFESTATIONS

Occurs weeks–months later:

 Myocarditis

 Pericarditis

 HIGH-YIELD FACT

 MOST COMMON CAUSE OF REVERSIBLE COMPLETE HEART BLOCK 

 Diagnosis

 Serology

 PCR

 Culture (slow & difficult)

 IV. CHAGAS DISEASE

 American Trypanosomiasis

 Causative Organism

 Trypanosoma cruzi

Flagellated protozoan parasite

 TRANSMISSION

Vector:

 RED  UV  IID BUG (TRIATOMINE BUG)

Also called:  “Kissing Bug” 

Transmission Mechanism

- 1 Bug bites during blood meal 
 - 2 Defecates near bite 
 - 3 Parasites enter through:  Scratching
-  Conjunctiva

PATHOGENESIS

Intracellular Multiplication

Parasites multiply as:

AMASTIGOTES

inside host cells

PSEUDOCYST FORMATION

Clusters of amastigotes form:

PSEUDOCYSTS

Tissue Destruction

Rupture causes:  Intense inflammation

Tissue destruction

AUTOIMMUNE INJURY

Parasitic antigens trigger:

SELF-DIRECTED IMMUNE RESPONSE

CLINICAL STAGES

1 ACUTE STAGE

Duration:

Acute Findings

 Fever

 Chagoma

 ROMAÑA SIGN

★ CLASSIC FINDING ★

 Unilateral eyelid edema

Occurs when entry is via:  Conjunctiva

 Blood Findings

 Trypomastigotes visible in blood

2 CHRONIC STAGE

Occurs decades later 

❤️ CARDIAC EFFECTS

🔥 Myocarditis

❤️ Chronic dilated cardiomyopathy

⚡ Arrhythmias

❤️ Heart failure

🍴 GASTROINTESTINAL EFFECTS

Parasites damage:  Myenteric plexus

💣 CONSEQUENCES

🍌 Megaesophagus

➡ Dysphagia

➡ Regurgitation

💣 Megacolon

➡ Severe constipation

→ Colonic dilation

DIAGNOSTIC TOOLS

 Acute Phase

 Direct microscopy: Detects trypomastigotes

 Chronic Phase

 Serology

 XENODIAGNOSIS

 UNIQUE EXAM FAVORITE

1 Starved triatomine bugs bite patient 

2 Feces examined after:

3 Parasites identified microscopically 

 PCR

Detects:  T. cruzi DNA

 HIGH-YIELD PEARLS ✨

☀️ Lemierre syndrome → Internal jugular vein 

☀️ Fusobacterium necrophorum causes Lemierre syndrome 

☀️ TB commonly causes caseous pericarditis 

☀️ Constrictive pericarditis = dreaded TB complication 

☀️ Tertiary syphilis → aortic root dilation 

☀️ Lyme carditis → reversible complete heart block ⚡

☀️ Chagas disease → dilated cardiomyopathy 

☀️ Romana sign = unilateral eyelid edema 

☀️ Megacolon & megaesophagus are classic GI findings 🍌

🎯 QUICK MEMORY TRICKS

🧠 LEMIERRE

“Lemi-Jugular” 💧 ➡️ Internal jugular thrombophlebitis

🧠 LYME CARDITIS

“Lyme stops the line” ⚡ ➡️ Reversible heart block

🧠 CHAGAS

“Chagas changes chambers & colon” ❤️ 🍌 ➡️ Dilated cardiomyopathy + megacolon

❤️ 🦠 INFECTIOUS DISEASES CAN DAMAGE THE HEART, VESSELS, VALVES, CONDUCTION SYSTEM & GI TRACT 🦠 ❤️